

SHAUNAK RANE

Associate AI Engineer • Agentic AI • LLM Systems • AI Automation

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PROFESSIONAL SUMMARY

AI Engineer with demonstrated experience designing and building autonomous multi-agent systems, structured output generation pipelines via LLM APIs, and production-ready RAG architectures. Proven track record deploying containerized REST microservices with FastAPI, PostgreSQL, and Docker, alongside predictive machine learning models in industrial time-series environments. Adept in prompt engineering, tool and function calling, vector retrieval workflows, and automated AI orchestration.

TECHNICAL SKILLS

AI & LLM Systems: Agentic AI, Multi-Agent Workflows, RAG, Prompt Engineering, Structured Outputs (JSON Schema), Tool/Function Calling, Vector Embeddings, LLM APIs (Gemini, OpenAI)

Machine Learning & Data: PyTorch, Scikit-learn, PyTorch Geometric, Statsmodels, Time-Series Modeling, Anomaly Detection, NumPy, Pandas

Backend & APIs: Python, FastAPI, Flask, REST APIs, WebSockets, SQL, TypeScript, C

Databases & DevOps: PostgreSQL, Supabase (pgvector), MySQL, Docker, Git, GitHub Actions, Postman, Linux, pytest

WORK EXPERIENCE

Univitt Technologies – AI Engineering Intern

May 2026 – Aug 2026 | Thane, India (Remote)

- Architected end-to-end predictive analytics pipeline for compressor Condition-Based Monitoring (CBM) across 3,879+ telemetry data points, validating 17/17 statistical quality gates.
- Engineered containerized REST inference microservices in Docker with Flask, automating anomaly detection and reducing operational diagnostic turnaround.
- Conducted automated feature selection and model benchmarking using Scikit-learn and Statsmodels, improving time-series sensor fault detection reliability.

Univitt Technologies – Software Engineering Intern

Apr 2025 – Jul 2025 | Thane, India (Remote)

- Developed high-throughput REST APIs using FastAPI and PostgreSQL for an enterprise food operations management platform, reducing endpoint query response latency.
- Implemented and deployed a Random Forest predictive attendance model into operational shift scheduling, attaining 88% forecast accuracy.
- Wrote automated unit test suites using pytest, managed database schema migrations, and collaborated on code reviews using Git and GitHub.

PROJECTS

Project Aegis – Autonomous Multi-Agent Triage System

Python, FastAPI, Agentic AI, RAG, Supabase (pgvector), Docker

- Architected an autonomous emergency response orchestrator with hybrid RAG retrieval and tool-calling agent swarms, enforcing strict JSON Schema validation for deterministic structured outputs.
- Engineered tool-calling agent workflows to autonomously query incident records, evaluate casualty severity, and dispatch prioritized emergency actions with sub-second API latency.
- Implemented semantic vector persistence via Supabase (pgvector) and structured audit trails for agent-to-agent message passing and real-time situational reporting.

Gridium Protocol – Multi-Agent Microgrid Optimization

Python, PyTorch, FastAPI, DDPG, WebSockets

- Developed a multi-agent reinforcement learning engine utilizing Deep Deterministic Policy Gradients (DDPG) to dynamically optimize power routing across a 15-node grid.
- Coordinated decentralized agent policies managing solar generation, battery storage, and dynamic loads, achieving an 18.4% reduction in peak-hour grid strain.
- Built real-time telemetry streaming pipelines using FastAPI and WebSockets, feeding continuous operational metrics to monitoring dashboards.

TopoFlow – GNN Fluid Dynamics Simulation

Python, PyTorch Geometric, GraphSAGE

- Formulated fluid velocity and pressure field modeling using Graph Neural Networks (GraphSAGE) over 1,231 mesh graphs, cutting MSE loss by 46.2% against baseline solvers.

EDUCATION

Universal AI University – B.Tech in Artificial Intelligence & Machine Learning

Expected May 2028 | Karjat, Maharashtra

- Key Coursework:** Data Structures & Algorithms, Machine Learning, Deep Learning, Database Management Systems (DBMS), Operating Systems, Linear Algebra.